

An energy method for the bifurcation analysis of necking

Yi Yan^{a,b}, Min Li^{b,c}, Zi-Long Zhao^d, Xi-Qiao Feng^{b,e,*}

^a Department of Engineering Mechanics, MIIT Key Laboratory of Dynamics and Control of Complex Systems, Northwestern Polytechnical University, Xi'an 710072, China

^b Institute of Biomechanics and Medical Engineering, AML, Department of Engineering Mechanics, Tsinghua University, Beijing 100084, China

^c College of Mechanics and Materials, Hohai University, Nanjing 211100, China

^d Institute of Solid Mechanics, School of Aeronautic Science and Engineering, Beihang University, Beijing 100191, China

^e State Key Lab of Tribology, Department of Engineering Mechanics, Tsinghua University, Beijing 100084, China

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ABSTRACT

Necking is an instability phenomenon widely observed in metallic and polymeric materials. On the basis of the minimum energy barrier criterion, we propose a theoretical method, combined with Monte Carlo simulations, to predict the linear instability and post-bifurcation behavior of materials with a non-convex constitutive law. For illustration, this method is applied to analyze the necking bifurcation behavior of a plate under uniaxial tension. Especially, we consider the effect of surface tension, which is important for the necking of polymeric materials. For soft materials with high surface energy, surface effects may greatly influence the morphological evolution of necking, such as the position and wavenumber of the consecutive necking zones. This work not only helps understand various necking phenomena in biological and engineering materials, but also can be extended to analyze some other instability problems, e.g., surface wrinkling.

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1. Introduction

Engineering structures under high mechanical loads may involve intriguing phenomena of instability, e.g., necking, shear localization, wrinkling, and cracking, which depend on the mechanical properties of materials, the shapes and characteristic sizes of structures, and loading conditions as well. For example, abrupt structural transitions such as buckling and wrinkling can occur in slender bars, thin plates, and shells under compressive loads. The classical Euler's theory of instability is often used to predict the onset of some bifurcation modes.

Under high loads, most structures have multiple possible equilibrium states, corresponding to different instability modes. In realistic situations, however, only one or a few instability modes are easily observed in a structure, depending on various influencing factors such as external perturbations, boundary constraints, and geometric defects. For instance, microstructural stochasticity or flaws may promote the formation of shear bands [1], material inhomogeneity may affect the onset of buckling and the path of crack propagation [2–4], strain rate may have a considerable impact on the necking behavior of metallic bars [5], and surface

wrinkling and morphological evolutions of biological tissues rely on the coupled mechanical–chemical mechanisms [6,7].

Necking of structures is an instability phenomenon that is closely related to the nonlinear constitutive relation of materials [8]. Ericksen discussed the instability of a one-dimensional elastic bar with a non-convex constitutive relation [9]. Recently, more complicated necking behaviors associated with such physical factors as electroelasticity [10] and surface tension [11] were reported. It has been found that a soft agar gel cylinder [12] and a hemispherical metallic shell [5] can be destabilized into multiple necks (Fig. 1). The morphological evolution of necking zones in a cylindrical bar can be experimentally observed by using a three-dimensional digital image correlation technique [13,14]. Recently, several theoretical models have been developed to investigate the necking behavior of polymer, hydrogel, and rubber-like soft materials. An earlier study of Leonov [15] demonstrated that the necking of polymers originates from the non-convex stress–strain relation. The non-convex mechanical behaviors of such soft materials as hydrogels are closely associated with damage evolution of their microscopic network structures [16,17]. Taffetani and Ciarletta [18] studied the effect of capillary energy on the beading instability of soft cylindrical gels. In these previous studies, however, the positions and wavenumber of the consecutive necking zones were usually predefined, and the underlying mechanisms of the complicated necking phenomena remain unclear [19]. In the present study, we make an attempt to understand the energetic mechanisms underlying the necking instability and predict the critical condition and morphological evolution of necking.

* Corresponding author at: Institute of Biomechanics and Medical Engineering, AML, Department of Engineering Mechanics, Tsinghua University, Beijing 100084, China.

E-mail address: fengxq@tsinghua.edu.cn (X.-Q. Feng).

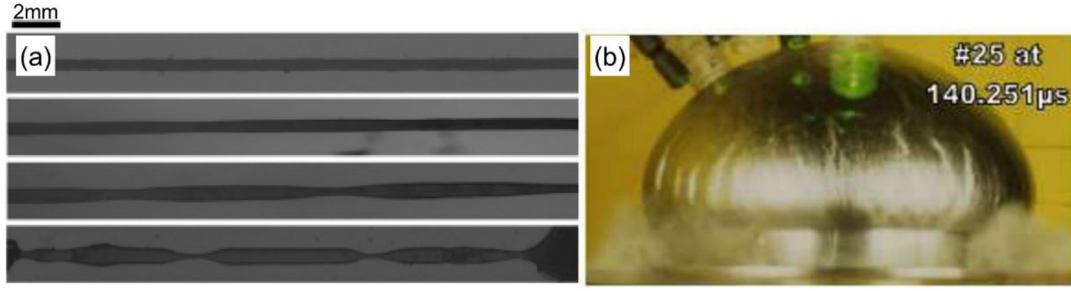


Fig. 1. Multiple necking modes in (a) a soft agar gel cylinder (from Mora et al. [12], used with permission from APS) and (b) a hemispherical metallic shell (from Mercier et al. [5], used with permission from Elsevier).

Energy principles provide an efficient tool to explore the physical mechanisms and critical conditions of various instability phenomena. In a pioneering work, Hill and Hutchinson [8] unraveled the unstable nature of necking, which is closely related to the convexity of the strain energy function. In the prediction of the critical condition of necking and the subsequent morphological evolution, it is of significance to choose an appropriate energy criterion. The classic Maxwell criterion [20] predicts the bifurcation of a system by using the minimum global potential energy. However, the bifurcation point predicted by the Maxwell criterion could be much earlier than the exact solution [21,22]. Bifurcation can be predicted by using the criterion of the minimum energy barrier and the concept of hysteresis [23]. In some cases, however, additional conditions or constraints are required by, for example, assuming that the post-bifurcation path has the steepest energy descent and the highest dissipation. In addition, it is emphasized that the bifurcation behavior of a structure can be affected by a number of external and internal factors. For example, surface energy may have a considerable influence on the onset of necking in polymeric and other soft materials [24]. To date, however, there is a lack of energy-based theoretical and numerical methods that can address the effects of such factors on the necking bifurcation behaviors.

We here present an energy method to investigate the linear instability and post-bifurcation behavior of materials with a non-convex constitutive relation. In particular, we consider the effect of surface tension on the necking of soft polymeric materials, which has been experimentally observed [24]. This paper is organized as follows. In Section 2, we describe the proposed energy method of necking analysis. In Section 3, the necking of a soft plate under uniaxial tension is taken as an example to demonstrate the efficiency of this method in predicting the necking behaviors, including deformation and energy evolution. Finally, the main conclusions drawn from this study are summarized.

2. Method

In this work, we propose an efficient method to investigate the necking problem on the basis of the principle of minimum potential energy. The necking process is tracked through a series of energy minimization procedures on the potential energy landscape of the system under loading. The proposed method is somewhat similar to the relaxation method in molecular dynamics simulations [25]. It enables us to analyze the necking behavior of materials with a non-convex constitutive relation, which is hard to be determined by the conventional theoretical methods.

2.1. Bifurcation criterion

To illustrate the basic idea of the proposed necking analysis method, we first consider a quasi-static two-dimensional (2D) plane strain problem. Let V and S represent the volume and

boundary of the body, respectively. The boundary is divided into two parts, $S = S_u \cup S_t$, where S_u and S_t denote the displacement- and traction-specified boundaries, respectively. Neglecting the body forces, the total potential energy in the system is expressed as

$$\Pi = W + P = \iint_V w[\boldsymbol{\varepsilon}(\mathbf{u})] dV - \int_{S_t} \mathbf{T} \cdot \mathbf{u} dS, \quad (1)$$

where $\mathbf{u}$ is the displacement vector, $\boldsymbol{\varepsilon}(\mathbf{u})$ is the strain tensor, $w[\boldsymbol{\varepsilon}(\mathbf{u})]$ is the elastic strain energy density, and $\mathbf{T}$ is the traction vector on the boundary S_t [26].

The equilibrium of the system requires that the variation of its total potential energy Π equals zero, that is,

$$\delta \Pi = \iint_V \delta w[\boldsymbol{\varepsilon}(\mathbf{u})] dV - \int_{S_t} \mathbf{T} \cdot \delta \mathbf{u} dS = 0. \quad (2)$$

Let $\mathbf{u}_{st}$ denote a displacement field that satisfies Eq. (2). The equilibrium condition in Eq. (2) is insufficient to ensure the stability of the state. For example, when $\mathbf{u}_{st}$ satisfies Eq. (2) but occupies a local energy maximum value or a saddle point, the state is unstable and would shift to a stable solution due to small perturbations. However, we can obtain a stable solution from Eq. (2) under certain conditions. In general, if the functional Π is strictly convex, the system has only one equilibrium state that occupies the global energy minimum [Fig. 2(a)]. For an elastic structure under a small load, one can obtain the stable solution from Eq. (2). With the increase in the external load, the system may have more than one equilibrium state. Fig. 2(b) shows such a characteristic shape of Π , where the new equilibrium states B and C have lower energy than the original stable state A [2]. Assume that the equilibrium states B and C have the same potential energy. Using the Maxwell criterion, we may predict two possible bifurcation modes, either from A to B or from A to C. Because of the energy barriers AD and AE, however, the equilibrium state A is also stable. In this case, the bifurcation predicted by the Maxwell criterion may be earlier than the actual bifurcation point [21].

According to the minimum energy barrier criterion, bifurcation occurs when the energy barrier between two equilibrium states disappears. Fig. 2(c) schematically shows a bifurcation mode with non-orientational path. If both the energy barriers AD and AE disappear simultaneously, the minimum energy barrier criterion predicts the occurrence of either the equilibrium state B or C. To explain the occurrence of a specific bifurcation mode, we need to consider such stochastic factors as external perturbations and internal structural randomness. Thereby, the potential energy landscape may transit to an orientational bifurcation mode, as illustrated in Fig. 2(d). The energy barrier AE tends to prevent the occurrence of bifurcation from A to C, and thus only the path from A to B may occur in the system. In what follows, we will show that the energy method proposed in this work will be able to predict such bifurcation modes.

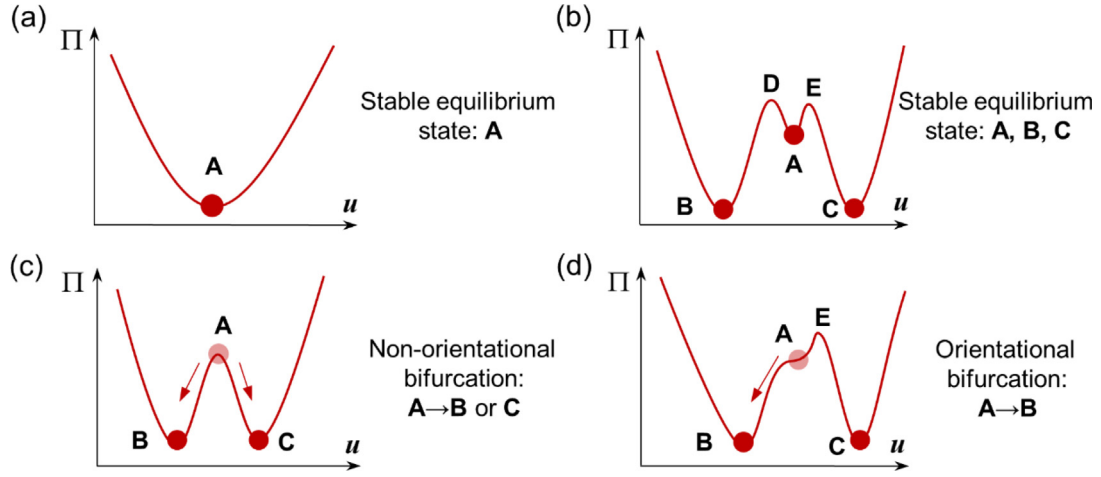


Fig. 2. Schematics of possible equilibrium states and the onset of bifurcation with orientational or non-orientational paths. (a) A stable system with a unique and stable equilibrium state, (b) a stable system with multiple equilibrium states, (c) an unstable system with non-orientational bifurcation, and (d) an unstable system with orientational bifurcation.

In this study, we will show that some physical mechanisms may trigger the transition from a non-orientational to an orientational bifurcation. Our method is proposed on the basis of the minimum energy barrier criterion. Assume that a solution bifurcation occurs from $\mathbf{u}'$ to $\mathbf{u}''$. There exists at least one continuous path of deformation that satisfies

$$\left. \frac{\delta \Pi}{\delta \mathbf{u}} \right|_{\mathbf{u}' \rightarrow \mathbf{u}''} \leq 0. \quad (3)$$

In the following sections, we will present both theoretical and numerical methods to solve the deformation path that satisfies Eq. (3).

2.2. Critical condition of necking

Now we use the proposed method to analyze a plate under uniaxial tension, as shown in Fig. 3(a). Let L and H denote its initial length and thickness, respectively. Assume that the width of the plate is much larger than the thickness, and thus the effect of width is neglected. For three-dimensional necking problems of plates with finite width, the present theoretical model still works but the finite element method is needed to solve the critical condition and morphological evolution of necking. Its boundary S contains four parts, i.e., $S = S_1 \cup S_2 \cup S_3 \cup S_4$. The boundaries S_1 and S_2 are specified with displacements as

$$\mathbf{u}^{S_1} = 0, \mathbf{u}^{S_2} = (\delta, 0)^T, \quad (4)$$

where δ denotes the tensile displacement applied at the right end. The boundaries S_3 and S_4 are traction-free, that is, $\mathbf{T}^{S_3} = \mathbf{T}^{S_4} = 0$. The equilibrium condition of the plate is written as

$$\delta \Pi = \iint_V \delta w[\boldsymbol{\varepsilon}(\mathbf{u})] dV = 0. \quad (5)$$

In this model, the displacement can be simplified as a scalar function

$$\mathbf{u}(x_1, x_2) \rightarrow u(x_1), \quad 0 \leq x_1 \leq L. \quad (6)$$

The boundary S of the model is expressed as

$$S = S_L \cup S_R, \quad (7)$$

and the displacement constraints reads

$$u^{S_L} = 0, u^{S_R} = \delta. \quad (8)$$

The equilibrium condition of the model requires that

$$\delta \Pi = H \int_0^L \delta w[u(x_1)] dx_1 = 0. \quad (9)$$

Here the potential energy is the sum of the elastic strain energy w_0 and the surface energy w_s :

$$W = H \int_0^L [w_0(\lambda) + w_s(\lambda)] dx_1, \quad (10)$$

where $\lambda = d[x_1 + u(x_1)]/dx_1$ is the elongation distribution in the plate. Besides, we define $\Lambda = (\delta + L)/L$ as the dimensionless displacement load.

Following Ref. [19], we use the following power-law form of the strain energy density w_0 :

$$w_0 = \frac{\sigma^*}{N+1} (\ln \lambda)^{N+1} + \alpha E_u \frac{(\lambda - \lambda_u)^2}{2\lambda_u^2}, \quad (11)$$

where σ^* , N , and α are constitutive parameters [Fig. 3(b)]. Eq. (11) can describe both hyperelastic and elastoplastic deformations of materials, which may exhibit different necking behaviors [19]. In the following, we will compare two limit cases that respectively have

$$\alpha = \begin{cases} 0, & \text{for elastic materials,} \\ 1, & \text{for elastic-plastic materials.} \end{cases} \quad (12)$$

For the latter case, let λ_u denote the maximum elongation at which the tensile force starts to decrease due to necking under the condition of displacement-controlled loading. For elastoplastic materials, E_u in Eq. (11) designates the unloading modulus, which may be different from the initial Young's modulus. In the following analysis, we set $E_u/\sigma^* = 300$.

In the initial stage of tension, the elongation λ is uniform along the tensile direction of the plate, i.e., $(0 \leq x_1 \leq L)$. This uniform solution is stable if it occupies the local potential energy minimum. From $d^2 w_0/d\lambda^2 = 0$, the critical load of bifurcation can be determined as

$$\lambda_{c0} = \exp(N). \quad (13)$$

Necking occurs when $\Lambda = \lambda_{c0}$ if the effect of surface energy w_s is negligible.

Surface effect may either delay or accelerate the onset of necking in soft plates, depending on the geometric shapes of the

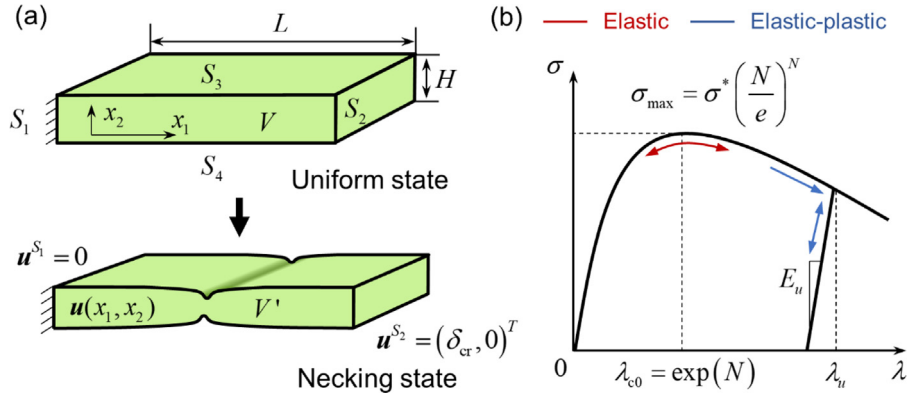


Fig. 3. (a) Schematics of the initial and the necked configurations of a soft plate under uniaxial tension. (b) Stress–elongation relation in the power-law form.

structure [24]. For the plane strain problem, the surface energy w_s is expressed as [24]

$$w_s = \frac{\gamma H^2}{2\lambda^5} \left(\frac{d\lambda}{dx_1} \right)^2 + \gamma\lambda, \quad (14)$$

where γ denotes the surface energy density. We neglect the surface energies of the end surfaces S_1 and S_2 (Fig. 3). In the uniform deformation mode, $d\lambda/dx_1 = 0$, Eq. (14) reduces to the uniform deformation solution prior to necking, and we have $w_s = \gamma\lambda$.

2.3. Theoretical model and numerical simulation

In most experiments, the necking instability of a slender bar follows an orientational bifurcation mode in the sense that the bar has only one necking zone, normally located at a position near the center. However, it is noticed that multiple necking zones may also appear at different positions in a structure of cylindrical, hemispherical, or other shapes [5,27]. In this study, we will analyze the physical mechanisms that lead to the occurrence of multiple necking morphologies. Surface effect on such orientational necking mode will be taken into account.

We invoke an energy-based method, using Eq. (3), to analyze the intrinsic orientation of necking. Assume that the deformation of the specimen bifurcates at a uniform elongation $\lambda(x_1) = \Lambda(\delta)$ under tensile displacement δ . The necking bifurcation mode is expressed as $\lambda(x_1) = \Lambda + \delta\lambda(x_1)$, where $\delta\lambda(x_1)$ is a deformation increment. Since $\delta\lambda(x_1)$ is much smaller than Λ in the early stage of necking, $\Lambda + \delta\lambda(x_1)$ covers the neighborhood around the uniform deformation Λ .

To satisfy the displacement constraint of the model, one has

$$\int_0^L \delta\lambda(x_1) dx_1 = 0. \quad (15)$$

By this method, we can predict not only the necking position but also the wavenumber of necking zones. To study the bifurcation mode of necking, $\delta\lambda(x_1)$ is assumed as

$$\delta\lambda(x_1) = A \cdot \sin\left(\frac{2\pi}{L} Kx_1\right), \quad (16)$$

where A represents the amplitude of the necking deformation increment, and K is an integer indicating the wavenumber of necking morphology. The value of A , which is much smaller than Λ , does not interfere with the major conclusions of this study.

Necking generally follows a long-wavelength instability mode, where the characteristic wavenumber could be much lower than 1 [28]. The sinusoidal function cannot well depict the necking at an arbitrary position in the specimen because the necking zone

($\delta\lambda > 0$) may emerge at one of its two ends. To guarantee $\delta\lambda(0) < 0$ and $\delta\lambda(L) < 0$ [19], a polynomial function is used to depict the necking displacement increment:

$$\delta\lambda(x_1) = \frac{B}{b} x_1^c (L - x_1)^{M-c} + d. \quad (17)$$

The three variables b , c , and d in Eq. (17) are expressed as

$$b = \frac{cL^c}{M} \left(L - \frac{cL}{M} \right)^{M-c}, \quad (18)$$

$$c = \frac{Mx_c}{L}, \quad (19)$$

$$d = \frac{-BL^M \Gamma(c+1) F(c+1, c-M; c+2, 1)}{b\Gamma(c+2)}. \quad (20)$$

where Γ is the Gamma function, and F the Gauss hypergeometric function. The shape of $\delta\lambda(x_1)$ in Eq. (17) mainly depends on three parameters: B , M , and x_c . B represents the amplitude of the necking displacement increment, M is the extent of strain localization, and x_c ($0 < x_1 < L$) refers to the necking position. For example, $x_c = L/2$ indicates the necking zone in the middle of the specimen.

To predict the necking wavenumber, we invoke the deformation increment $\delta\lambda(x_1)$ of Eq. (16) with specified parameters (A, K). Thereafter the necking position can be determined by using the deformation increment $\delta\lambda(x_1)$ expressed in Eq. (17) under a group of given parameters (B, M, x_c). The influences of these parameters on $\delta\lambda(x_1)$ are illustrated in Fig. 4(a) and (b). Once $\delta\lambda(x_1)$ has been determined, we can derive the critical elongation Λ_c of the necking mode $\Lambda + \delta\lambda(x_1)$ through

$$\Delta\Pi = \Pi(\Lambda_c + \delta\lambda) - \Pi(\Lambda_c) = 0. \quad (21)$$

From Eq. (21), we can determine the critical load Λ_c for a specific necking mode. Fig. 4(c) schematically shows several typical morphological evolutions of necking and the corresponding critical loads. Each necking mode has a critical load, and thus a continuous curve of Λ_c forms corresponding to all possible necking modes. According to the minimum energy barrier criterion, the actual necking mode corresponds to the lowest critical load, and all other bifurcation modes have higher energy barriers.

It is mentioned that the theoretical method considers only the neighborhood of the uniform deformation, i.e., the regime of linear stability analysis. In the next section, we will investigate, through numerical simulations in the nonlinear post-bifurcation regime, the effect of surface tension on the entire evolution process of necking morphology.

The Monte Carlo (MC) method is extensively used to study the randomness in the geometric [29] and material properties [30–32] of a continuum with complex microstructures. By taking the

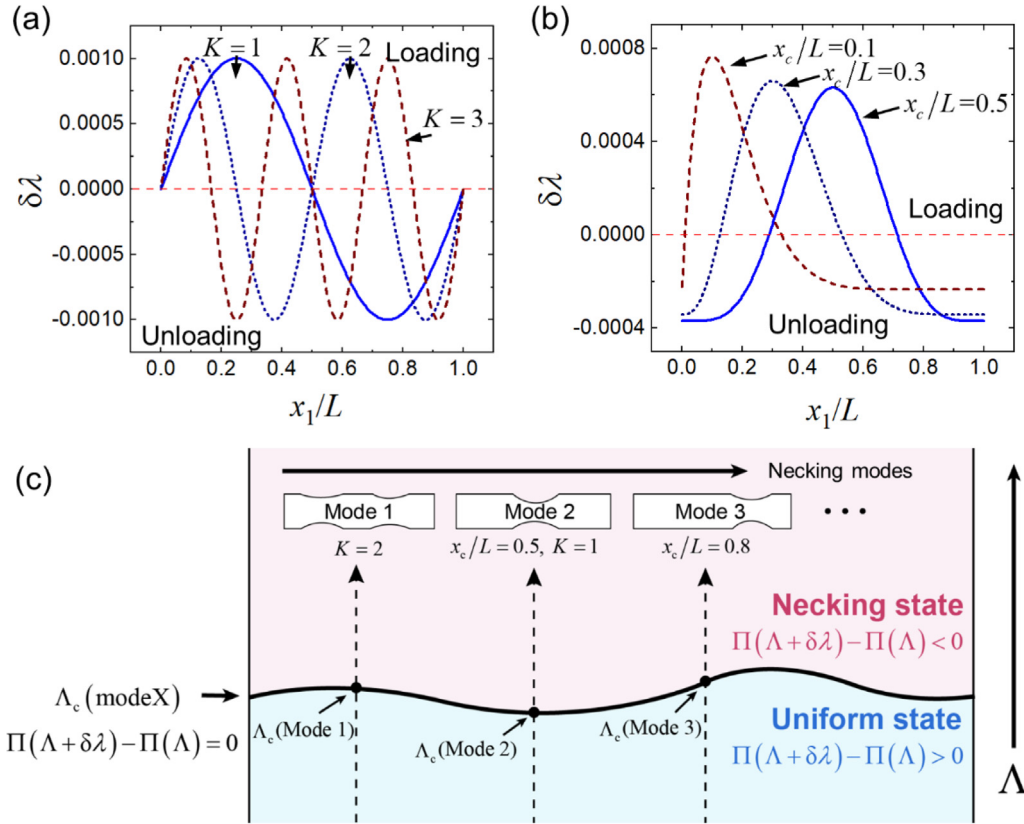


Fig. 4. Shape of the necking displacement increment $\delta\lambda(x_1)$. (a) The sinusoidal distributions for the necking wavenumber analysis, and (b) the polynomial distributions for the necking position analysis. (c) Schematic of the theoretical necking analysis. Three characteristic morphologic evolutions of necking with different bifurcation points Λ_c are shown as examples.

displacement field $\mathbf{u}$ as a random variable, the potential energy of a discrete system can be determined. MC simulations have also been used in investigations of small-scale particle systems [33–36]. Based on the minimum energy barrier criterion, we convert Eq. (3) to an optimization problem and solve it through MC simulation. By introducing numerical stochasticity to the system, the MC simulation could determine all necking modes that satisfy Eq. (3).

The MC method has two advantages in solving the stable equilibrium state for a system with a non-convex constitutive law. First, the random deformation assumed in MC simulations may trigger the bifurcation during the energy minimization process, and thus no additional disturbance is needed. Second, the MC algorithm is able to predict the physically possible bifurcation mode without imposing any constraints. See the Appendix for the detailed algorithm.

3. Results and discussions

In this section, we first present the results of MC simulations, including the energy descending process and necking morphological evolutions. Then the theoretical analysis and MC simulation are combined to examine the surface effect on the necking bifurcation and post-bifurcation behaviors. All parameters used in the theoretical analyses and numerical simulations are given in Table 1.

3.1. Necking modes

The proposed MC method, which does not require imposing any constraints, can well reproduce various necking phenomena and reveal the underlying energetic mechanisms. Through MC

Table 1

Parameters used in the theoretical and numerical analyses.

Material parameters			Geometric parameters	
σ^*	N	γ	H	L
1.0	0.3	0.02	0.5	2.0
Theoretical analysis				
A	B	M	K	x_c/L
0.001	0.001	10.0	1–5	0.1–0.9
MC simulations				
m	Number of trials			
20	20 000			

simulations, we find that there exist two typical necking modes, stable and unstable [19], as shown in Fig. 5. Fig. 5(a) and (b) show that the unstable necking mode stems from the elastic constitutive law we have assumed in Eq. (11) with $\alpha = 0$. In this situation, strain localization rapidly occurs at the position of necking with the increase of displacement loading. Thus, the unstable necking deformation will soon lead to the fracture of the plate. Different from the uniform deformation stage before necking, the necking plate evolves into three parts, including the necking or shrinking part and two elastic-unloading parts [Fig. 5(b)].

The stable necking mode may happen in elastoplastic materials, as shown in Fig. 5(c) and (d). In this case, after the occurrence of necking, the tensile force increases with the externally applied displacement loading. When the neck shrinks to a certain thickness, the necking deformation will become unstable and finally lead to the fracture of the plate. This finding is consistent with the results of Audoly and Hutchinson [19].

Define the energy descending ratio as

$$\Pi_{\text{descend}} = \frac{\Pi_{\text{MC}} - \Pi_{\infty}}{\Pi_{\infty}}, \quad (22)$$

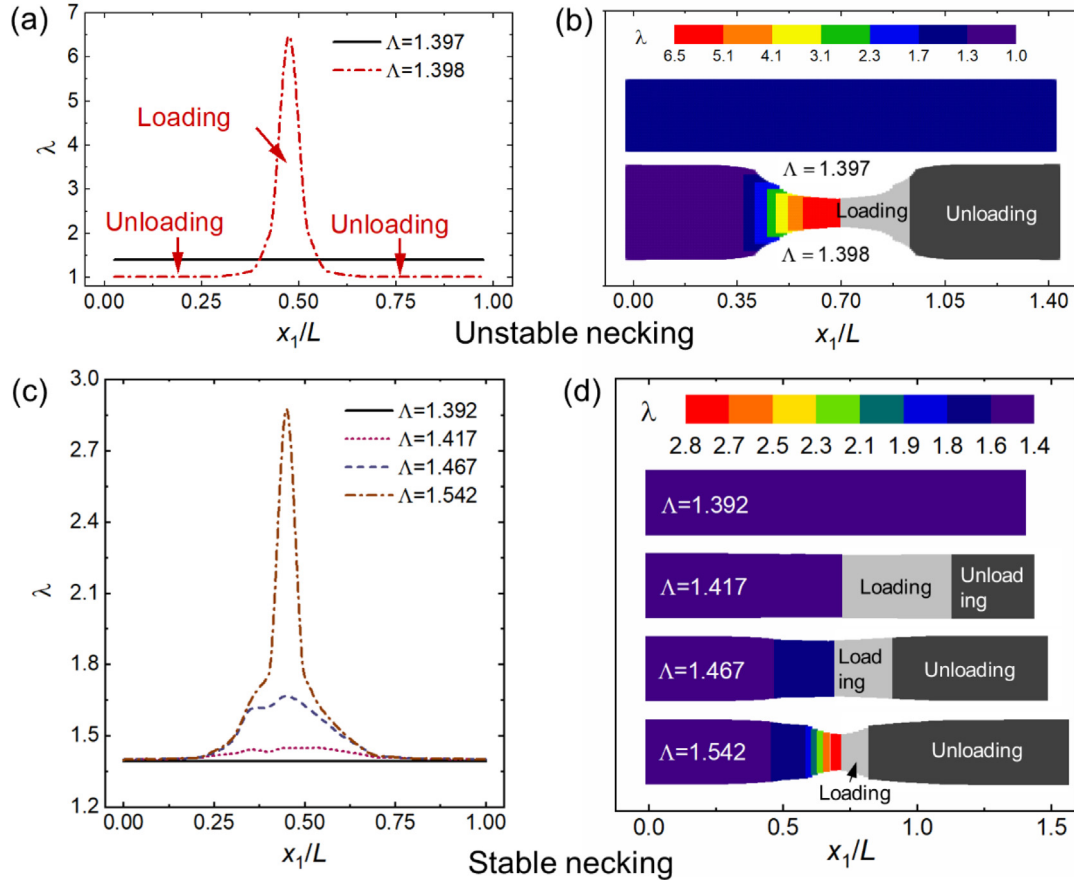


Fig. 5. Unstable necking (a, b) and stable necking (c, d) predicted by MC simulations. (a, c) The distribution of stretch λ in the specimen along the tensile direction, and (b, d) the equivalent two-dimensional necking deformation. In the necking configuration, the colors on the left half-length indicate the distribution of stretch λ , while the grayscale on the right side shows the loading and unloading parts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

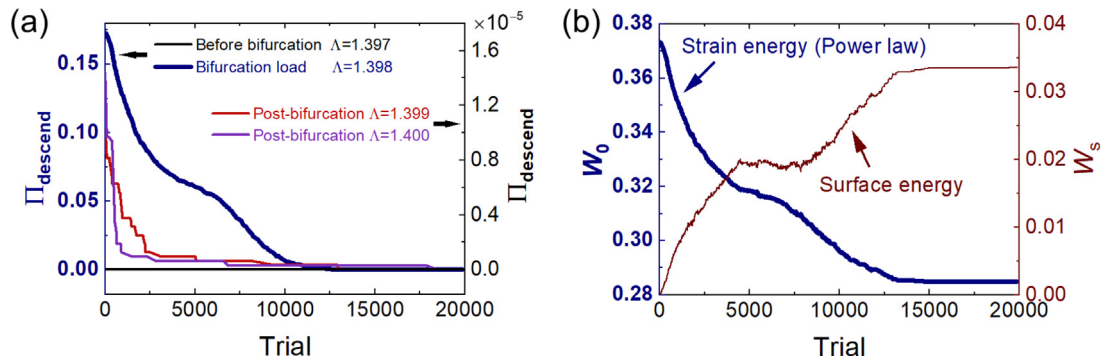


Fig. 6. Energy descending curves of the unstable necking mode obtained from MC simulations. (a) The energy descending ratios before (black), at (cyan bold), and after (red and purple) the critical load $\Lambda_c = 1.398$ where necking occurs, and (b) optimization of the elastic strain energy (cyan bold) and surface energy (brown) at the critical load $\Lambda_c = 1.398$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

where Π_{MC} denotes the potential energy during the MC energy minimization process. Π_{∞} is the energy of the last accepted MC trial, which is considered as the energy at the stable equilibrium state.

For each displacement load Λ , the MC simulation of energy minimization converges after about 10,000 trials. Fig. 6(a) shows the energy minimization processes of the unstable necking mode from $\Lambda = 1.397$ to 1.400 , where $\Lambda = 1.398$ is the critical load. For the load steps before the occurrence of necking, no energy descending is observed because the uniform initial configuration is a stable solution in this regime. A sudden drop of energy is

observed at the critical load $\Lambda_c = 1.398$. Thereafter, energy descending is induced by the incremental displacement load. Fig. 6(b) shows the evolution of both the elastic strain energy and surface energy at the critical load $\Lambda_c = 1.398$. The strain energy W_0 decreases while the surface energy W_s increases. It is emphasized that, in this example, though W_s is one order magnitude smaller than W_0 , surface energy may significantly affect the bifurcation mode.

For the stable necking mode, the total potential energy increases further in the early stage of necking, and it starts to decrease when the externally applied displacement reaches the

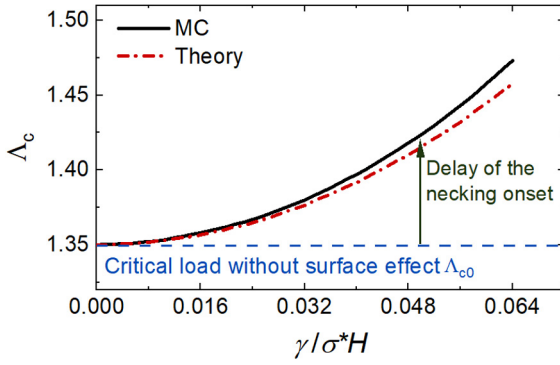


Fig. 7. Variation of the critical necking displacement load Λ_c with the dimensionless surface energy $\gamma/(\sigma^*H)$, where Λ_{c0} denotes the critical stretch without surface effect [$\gamma/(\sigma^*H) = 0$].

critical load $\Lambda_c = 1.398$. In contrast to the unstable necking mode, the variation in the total potential energy is more smooth during the necking process.

3.2. Surface effect on the necking critical condition

Next, we use the method presented in Section 2 to investigate the effect of surface tension on the necking behavior, characterized by the dimensionless parameter $\gamma/(\sigma^*H)$. Fig. 7 gives the critical displacement loads of necking, Λ_c , obtained from the theoretical analysis and MC simulation. Clearly, the critical load Λ_c shows a strong dependence on the surface energy. Λ_c increases with the increase in $\gamma/(\sigma^*H)$, suggesting that the surface energy tends to delay the onset of necking in a plate, which is consistent with the previous study [19]. The delay is induced by the increase of surface energy with the evolution of necking. When the surface effect is neglected [$\gamma/(\sigma^*H) = 0$], the critical load reduces to the classical solution $\Lambda_c = \exp(N)$.

3.3. Surface effect on the necking position

Besides the critical load of necking, we investigate the surface effect on the necking morphology, which can be predicted by the theory in Section 2. The phase diagrams of a plate are shown in Fig. 8, where the total potential energy change $\Delta\Gamma$ is given as a function of the mechanical stretch Λ and the necking position x_c . The solid curve divides the phase diagram into the uniform deformation state ($\Delta\Gamma > 0$) and the necking state ($\Delta\Gamma < 0$). The interface between the two states corresponds to the critical load Λ_c with respect to various possible necking positions x_c . The smallest value of Λ_c corresponds to the necking mode that will really occur.

When the effect of surface tension is considered, the necking position predicted by the theoretical analysis is shown in Fig. 8(a), where we take the normalized surface energy density $\gamma/(\sigma^*H) = 0.04$. Among all possible necking positions (e.g., $x_c/L = 0.1, 0.2, 0.3, 0.4$, and 0.5), the critical necking stretch Λ_c takes the minimal value when necking occurs at the middle of the plate ($x_c/L = 0.5$). Therefore, due to the surface effect, the minimum energy barrier criterion is satisfied only at the necking position of $x_c/L = 0.5$.

For comparison, when the surface effect is neglected [i.e., $\gamma/(\sigma^*H) = 0$], the necking position predicted by the theoretical analysis is shown in Fig. 8(b). It is seen that the critical stretch of necking is a constant, $\Lambda_c = 1.350$, independent of the necking positions. Therefore, in this case, necking may occur at any position in the plate or, in other words, in a non-orientational bifurcation fashion. The real necking position would be affected

by such factors as geometric defects in the specimen. The above results show that due to the effect of surface tension, the bifurcation mode of necking position turns from a non-orientational [$\gamma/(\sigma^*H) = 0$] to an orientational mode [$\gamma/(\sigma^*H) > 0$].

We now compare the results from the theoretical analysis and MC simulations. We perform parallel simulations of 500 samples in each case [$\gamma/(\sigma^*H) = 0, 0.025$, and 0.04]. The possibility density distributions for necking positions are shown in Fig. 9. It is seen that the effect of surface tension can also be reflected by the possibility density distributions obtained from the MC simulations (Fig. 9). In the absence of surface energy [$\gamma/(\sigma^*H) = 0$], corresponding to the non-orientational bifurcation, the possibility density distribution for necking positions is relatively uniform. When $\gamma/(\sigma^*H) = 0.04$, the possibility density distribution has a much higher peak in the middle of the specimen. This result is identical to the theoretical analysis, though the latter method predicts a determinant necking position at $x_c/L = 0.5$. The random distribution of necking positions obtained from the MC simulations results from the stochastic nature of this method, and it results from the perturbations or fluctuations in real structures, and the bifurcation mode with a minor energy barrier can be tracked.

3.4. Surface effects on the necking wavenumber

Besides the necking position, another important but unsolved issue is the prediction of the necking wavenumber when necking occurs at multiple sites in a specimen, as shown in Fig. 1 [5,12]. Now we use the proposed method in Section 2 to investigate the wavenumber of multiple necking positions with surface effect.

Fig. 10(a) plots the total potential energy and its relative change $\Delta\Gamma$ as a function of the mechanical stretch Λ and the necking mode K when the effect of surface tension is considered by taking $\gamma/(\sigma^*H) = 0.04$. Let us assume that the plate under uniaxial tension would destabilize into a necking morphology with wavenumber K . The total potential energy changes for the first five necking modes ($K = 1 - 5$) are compared in Fig. 10. Obviously, the critical necking stretch Λ_c at $K = 1$ takes the minimum value among all possible necking modes. In general, therefore, only one necking zone is observable in plates under quasi-static uniaxial tension. The surface energy tends to hinder the occurrence of higher-order necking modes.

However, when the surface effect is neglected (i.e., $\gamma/(\sigma^*H) = 0$), the critical stretch of necking is constant, $\Lambda_c = 1.350$, independent of the necking wavenumber, as shown in Fig. 10(b). In this case, all necking modes have the same critical stretch, and therefore necking occurs in a non-orientational bifurcation fashion. For this reason, on the basis of the minimum energy barrier criterion, the classical necking model without surface effect cannot explain the occurrence of a certain necking wavenumber, and the presence of surface energy makes the necking of plates as an orientational bifurcation mode with $K = 1$. If other physical mechanisms are considered, such as the strain rate, the above finding may help understand the experimental observation of multiple necking sites [37].

4. Conclusions

In summary, we have presented an energy method to investigate the bifurcation behavior of necking by combining the theoretical method with Monte Carlo simulations. Without imposing any constraints on the bifurcation mode, this method is able to predict the critical condition and post-bifurcation behavior of necking. It is found that for soft materials with high surface energy, surface effects play a crucial role in the morphological evolution of necking, such as the position and wavenumber of

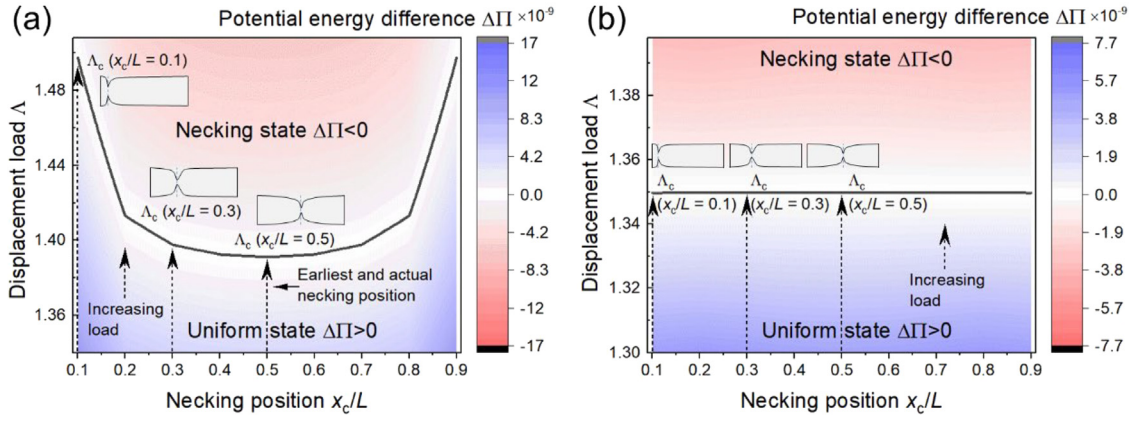


Fig. 8. Phase diagrams of the theoretically predicted necking position: (a) surface energy density $\gamma/(\sigma^*H) = 0.04$, and (b) $\gamma/(\sigma^*H) = 0$. The colors indicate the potential energy difference $\Delta\Pi$ in the total potential energy with respect to that at the uniform deformation state when the necking occurs at position x_c . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

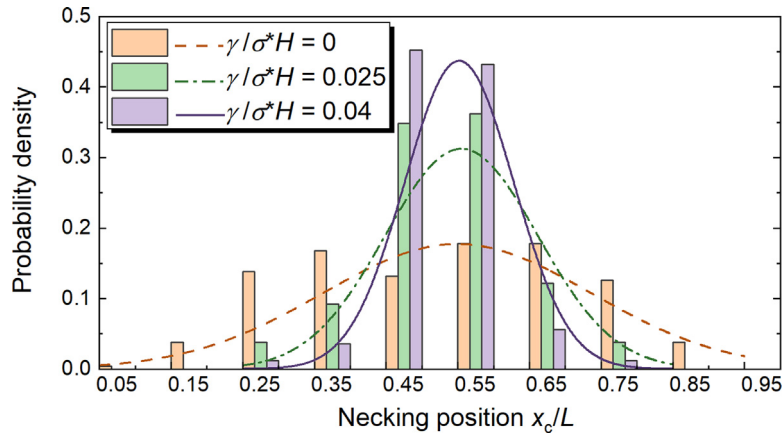


Fig. 9. Probability density distributions for necking positions obtained from the MC simulations, where 500 samples are calculated in each case. The bold curves denote the regressed Gauss distribution. The maximal probability of necking at the middle of the specimen increases with increasing $\gamma/(\sigma^*H)$.

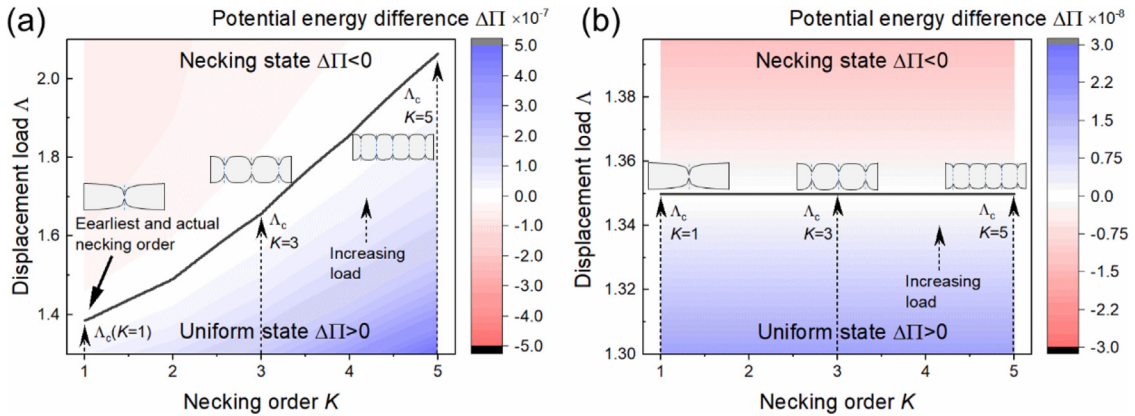


Fig. 10. Necking mode predicted by the theoretical analysis: (a) surface energy density $\gamma/(\sigma^*H) = 0.04$, and (b) $\gamma/(\sigma^*H) = 0$. The colors indicate the potential energy difference $\Delta\Pi$ in the total potential energy with respect to that at the uniform deformation state when the necking occurs at the wavenumber K . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the consecutive necking zones. For example, a plate with higher surface tension under uniaxial tension will destabilize into a morphology with only one neck in the middle region, but it may evolve into high-order necking modes with multiple necks if its surface energy is negligible. Due to these mechanisms, the surface effect may also affect the load-bearing capacity of soft polymeric materials. In general, surface tension tends to delay the

occurrence of necking and thus increase the load-bearing capacity of plates. The results presented in this letter can well reproduce a diversity of experimentally observed necking phenomena. This work not only gains insights into the physical mechanisms underlying various necking phenomena in engineering structures, but also helps design soft devices and systems with enhanced performance.

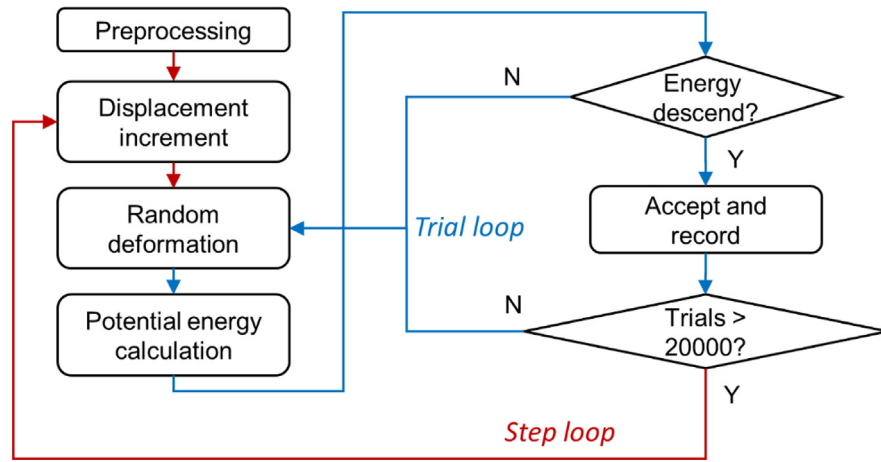


Fig. A.1. Flowchart of the MC simulation method, where the blue and red arrows denote the trial and step loops, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Finally, it is emphasized that there exist other factors that may affect the necking behavior of engineering structures, such as surface or internal flaws, strain rate, and viscous properties. Besides, the proposed method can also be extended to investigate other instability phenomena such as surface wrinkling and creasing in soft materials [38].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Algorithm of Monte Carlo simulation

We perform Monte Carlo simulations to predict the formation of necking by using the random energy minimization algorithm. The Monte Carlo method differs from the theoretical analysis in two main aspects. First, the bifurcation mode is selected automatically in the MC simulations, while the theoretical analysis prescribes one or more specific bifurcation modes. Second, the MC method can track the complete bifurcation and post-bifurcation process of the structure under loading, including the post-bifurcation path with the minimal energy barrier.

The two-dimensional plate S is discretized into a node–element system, consisting of m elements. The displacement field is expressed as $u_i(x_i)$ ($i = 0, 1, \dots, m+1$), where x_i denotes the coordinate of the node i . The elongation is expressed as $\lambda_i(x_i) = m(u_{i+1} - u_i)/L$ ($i = 0, 1, \dots, m-1$). The total potential energy is written as

$$\Pi(u_0, u_1, \dots, u_{m+1}) = \sum_{i=0}^{m-1} [w_{0i}(\lambda_i) + w_{si}(\lambda_i)], \quad (\text{A.1})$$

where w_{0i} and w_{si} denote the elastic strain energy and surface energy in the element i , respectively.

Then we perform MC simulations to solve the following energy minimization problem

$$\begin{aligned} \min \Pi(u_0, u_1, \dots, u_{m+1}), \\ \text{subject to: } u_0 = 0, u_{m+1} = \delta. \end{aligned} \quad (\text{A.2})$$

Fig. A.1 shows the flow chart of the MC method for necking analysis, which comprises two loops. The energy minimization is achieved in the trial loop, and the non-convex potential energy landscape is updated in the step loop.

References

- [1] W. Chen, Z. Liu, H.M. Robinson, J. Schroers, *Acta Mater.* 73 (2014) 259–274.
- [2] Y. Yan, A. Nakatani, *J. Mech. Phys. Solids* 125 (2019) 591–612.
- [3] V. Slesarenko, S. Rudykh, *J. Mech. Phys. Solids* 99 (2017) 471–482.
- [4] N. Arora, A. Batan, J. Li, V. Slesarenko, S. Rudykh, *Materials* 12 (2019) 763.
- [5] S. Mercier, N. Granier, A. Molinari, F. Llorca, F. Buy, *J. Mech. Phys. Solids* 58 (2010) 955–982.
- [6] B. Li, Y.P. Cao, X.Q. Feng, H. Gao, *Soft Matter* 8 (2012) 5728–5745.
- [7] S.L. Xue, S.F. Yin, B. Li, X.Q. Feng, *J. Mech. Phys. Solids* 121 (2018) 463–479.
- [8] R. Hill, J.W. Hutchinson, *J. Mech. Phys. Solids* 23 (1975) 239–264.
- [9] J.L. Ericksen, *J. Elasticity* 5 (1975) 191–201.
- [10] Y. Fu, L. Dorfmann, Y. Xie, *Extreme Mech. Lett.* 21 (2018) 44–48.
- [11] C. Xuan, J. Biggins, *Phys. Rev. E* 94 (2016) 1–7.
- [12] S. Mora, T. Phou, J.-M. Fromental, L.M. Pismen, Y. Pomeau, *Phys. Rev. Lett.* 105 (2010) 214301.
- [13] S. Ma, Z.L. Zhao, X. Wang, *J. Strain Anal. Eng. Des.* 47 (2012) 163–175.
- [14] P.D. Versailliot, Y.F. Wu, Z.L. Zhao, *Int. J. Mech. Sci.* 189 (2021) 106002.
- [15] A.I. Leonov, *Int. J. Solids Struct.* 39 (2002) 5913–5926.
- [16] Y. Liu, H. Zhang, Y. Zheng, *J. Appl. Mech.* 83 (2016) 1–14.
- [17] V. Morovati, M.A. Saadat, R. Dargazany, *Phys. Rev. E* 102 (2020) 062501.
- [18] M. Taffetani, P. Ciarletta, *J. Mech. Phys. Solids* 81 (2015) 91–120.
- [19] B. Audoly, J.W. Hutchinson, *J. Mech. Phys. Solids* 97 (2016) 68–91.
- [20] I. Müller, *Arch. Ration. Mech. Anal.* 70 (1979) 61–77.
- [21] G. Puglisi, L. Truskinovsky, *J. Mech. Phys. Solids* 50 (2002) 165–187.
- [22] Y. Yan, A. Nakatani, *J. Biomech.* 86 (2019) 183–192.
- [23] I. Müller, P. Villaggio, *Arch. Ration. Mech. Anal.* 65 (1977) 25–46.
- [24] M. Li, Y. Yan, S. Xu, G. Wang, J. Wu, X.Q. Feng, *Curr. Appl. Phys.* 38 (2022) 91–98.
- [25] D. Frenkel, B. Smit, *Understanding Molecular Simulation: From Algorithms to Applications*, Elsevier, Orlando, USA, 1996.
- [26] A.F. Bower, *Applied Mechanics of Solids*, CRC Press, New York, 2009.
- [27] K.E. N'Souglo, A. Srivastava, S. Osovski, J.A. Rodríguez-Martínez, *Proc. Math. Phys. Eng. Sci.* 474 (2018) 20170575.
- [28] Y. Fu, L. Jin, A. Gorieli, *J. Mech. Phys. Solids* 147 (2021) 104250.
- [29] X. Wang, Z. Yang, A.P. Jivkov, *Constr. Build. Mater.* 80 (2015) 262–272.
- [30] Z. Yang, X.F. Xu, *Comput. Methods Appl. Mech.* 197 (2008) 4027–4039.
- [31] X.T. Su, Z.J. Yang, G.H. Liu, *Int. J. Solids Struct.* 47 (2010) 2336–2345.
- [32] P.F. Liu, J.Y. Zheng, *Mater. Sci. Eng. A* 425 (2006) 260–267.
- [33] A. Puzder, A.J. Williamson, F.A. Reboredo, G. Galli, *Phys. Rev. Lett.* 91 (2003) 1–4.
- [34] J. Ma, D. Alfè, A. Michaelides, E. Wang, *Phys. Rev. B* 80 (2009) 1–4.
- [35] E.H. Kellogg, A. Leaver-Fay, D. Baker, *Proteins Struct. Funct. Genet.* 79 (2011) 830–838.
- [36] P.D. Spanos, A. Kontsos, *Probab. Eng. Mech.* 23 (2008) 456–470.
- [37] B. Audoly, J.W. Hutchinson, *J. Mech. Phys. Solids* 123 (2019) 149–171.
- [38] X.Q. Feng, Y.P. Cao, B. Li, *Surface Wrinkling Mechanics of Soft Materials*, Science China Press, Beijing, 2017.